

Sara Ghazanfari

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ABOUT ME

PhD candidate at NYU ECE (expected Dec. 2026) researching multimodal large language models, with a focus on video understanding, cross-modal alignment, and systematic evaluation of foundation models. First-author publications at CVPR, ICLR, and TMLR spanning synthetic benchmark construction, parameter-efficient architecture design, and post-training of vision-language models. Research intern at Apple and Adobe, bridging state-of-the-art research with applied systems.

EXPERIENCE

- **Applied Research Scientist Intern, Apple Inc.**, Cupertino, US May. 2026 – Aug. 2026
 - Designed a lightweight video temporal grounding system for an internal visual search product, achieving superior accuracy over state-of-the-art foundation models at substantially reduced model size, with improved retrieval accuracy, inference speed, and storage efficiency over the production baseline.
- **GenAI Research Intern, Adobe Inc.**, San Jose, US May. 2025 – Aug. 2025
 - Released SpotEdit (NeurIPS W 2025), a benchmark for visually-guided image editing that introduces the first dedicated hallucination evaluation subset — exposing systematic failure modes in GPT-4o and other state-of-the-art models when visual reference is absent.
 - Benchmarked leading models (OmniGen2, BAGEL, GPT-4o), identifying distinct capability profiles: OmniGen2 excelled at visual instruction following while BAGEL showed stronger background fidelity.
- **Research Assistant, New York University (NYU)**, NY, US Jan. 2023 – Dec. 2026
 - Published first-author research on multimodal reasoning and video understanding (CVPR, ICLR, TMLR), advancing temporal, spatial, and physical reasoning in vision-language models.
 - Developed post-training and parameter-efficient fine-tuning methods (instruction tuning, LoRA, adapter-based cross-modal alignment) for vision-language models across multimodal LLM and video understanding tasks.
 - Built synthetic benchmarks and large-scale evaluation pipelines (Habitat, Kubric, CLEVRER) to systematically probe multimodal reasoning gaps across 22 state-of-the-art MLLMs.
 - Designed evaluation frameworks for temporal, spatial, and physical reasoning, incorporating human baselines, bootstrap confidence intervals, and sim-to-real correlation analyses.
- **Technical Team Lead**, Narvan Startup Studio, Iran Oct. 2021 – Jan. 2023
 - Led a 6-engineer team to design and ship a full-stack cryptocurrency exchange platform (web + mobile), architecting a scalable backend with PostgreSQL, Redis, and Kafka.
- **Data Engineer**, Narvan Startup Studio, Iran Aug. 2020 – Oct. 2021
 - Built real-time streaming pipelines (Kafka, PostgreSQL, Elasticsearch, InfluxDB, Redis) and developed an end-to-end NLP pipeline for Persian news classification, improving topic accuracy by 15%.

EDUCATION

Ph.D. Candidate Jan. 2023 – Dec. 2026

Electrical and Computer Engineering, New York University (NYU)

GPA: 3.9/4.0

Advisors: Siddharth Garg, Farshad Khorrani

Research Area: Multimodal Large Language Models (MLLMs), Vision-Language Models (VLMs), Foundation Models, Generative AI, Computer Vision, Deep Learning

Courses: Probability & Stochastic Processes (A), Deep Learning (A), Algorithmic Machine Learning & Data Science (A⁺), Linear Systems (A), Machine Learning for Cybersecurity (A).

Master of Science

Sep. 2018 – Jul. 2021

Computer Engineering, Sharif University of Technology (SUT)

GPA: 4.0/4.0

Thesis: Isoform Function Prediction Using a Deep Neural Network

<u>SYNCR</u>: Diagnosing and Learning Cross-Video Reasoning from Simulation	Preprint 2026
<u>SYNCR</u>: A Cross-Video Reasoning Benchmark with Synthetic Grounding	NeurIPS W 2026
<u>Chain-of-Frames</u>: Advancing Video Understanding in Multimodal LLMs	CVPR 2026
<u>SpotEdit</u>: Evaluating Visually-Guided Image Editing Methods	NeurIPS W 2025
<u>Towards Unified Benchmark and Models for Multi-Modal Perceptual Metrics</u>	CVPR W 2025
<u>EMMA</u>: Efficient Visual Alignment in Multimodal LLMs	TMLR 2025
<u>LipSim</u>: A Provably Robust Perceptual Similarity Metric	ICLR 2024
<u>R-LPIPS</u>: An Adversarially Robust Perceptual Similarity Metric	ICML W 2023

SELECTED PROJECTS

🔗 SYNCR: Diagnosing and Learning Cross-Video Reasoning from Simulation

A simulator-grounded framework (Habitat, Kubric, CLEVRER) for diagnosing and learning cross-video reasoning, with 4,000 evaluation and 15,960 training questions across eight tasks. Evaluating 22 MLLMs reveals persistent failures; fine-tuning on SYNCR nearly doubles accuracy and transfers to real footage.

🔗 Chain-of-Frames (CoF): Advancing Video Understanding in Multimodal LLMs

A frame-aware chain-of-thought reasoning strategy for video LLMs that grounds each reasoning step in explicit frame references throughout the inference process. Improves average accuracy by 5.1 points across five video understanding benchmarks, with gains up to +10.3 points on VSI-Bench over standard chain-of-thought baselines.

🔗 SpotEdit: Evaluating Visually-Guided Image Editing Methods

A benchmark for visually-guided image editing across diffusion, autoregressive, and hybrid architectures, introducing the first dedicated hallucination evaluation subset. Exposes systematic failure modes in GPT-4o and other state-of-the-art models when visual reference replaces text instruction.

🔗 EMMA: Efficient Visual Alignment in Multimodal LLMs

A lightweight cross-modal alignment module (<0.2% parameter overhead) for multimodal LLMs, generating instruction-aware visual representations via CLIP’s text encoder. Achieves state-of-the-art on 7 of 10 benchmarks — up to +9.3% on MMVP and +5.6% on MuirBench — with a module 50× smaller than prior work.

🔗 LipSim: A Provably Robust Perceptual Similarity Metric

A provably robust perceptual similarity metric built on 1-Lipschitz neural networks with certified guarantees against adversarial attacks. Addresses demonstrated vulnerabilities in LPIPS while maintaining competitive natural performance via knowledge distillation from ViT-based models.

SKILLS

Research Expertise: Large Language Models (LLMs), Vision-Language Models (VLMs), Foundation Models, Video Understanding, Multimodal Reasoning, Generative AI, Diffusion Models, Autoregressive Models, Transformer Architectures, Cross-modal Alignment, Fine-tuning, Instruction Tuning, Post-training, Parameter-Efficient Fine-Tuning (PEFT), Knowledge Distillation, Adversarial Robustness, Benchmark Design, Synthetic Data Generation, Human Evaluation Design, Natural Language Processing (NLP).

Programming: Python, C.

ML Frameworks & Tools: PyTorch, HuggingFace Transformers, HuggingFace Accelerate, DeepSpeed, FSDP, vLLM, scikit-learn, spaCy, Weights & Biases (wandb).

Infrastructure & Simulation: CUDA, SLURM/HPC, Docker, Git/GitHub, Unix Shell, Blender, PyBullet.

Data Infrastructure: PostgreSQL, Redis, Elasticsearch, InfluxDB, Kafka.

AWARDS & HONORS

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| • School of Engineering (SOE) Fellowship (merit-based doctoral fellowship), NYU ECE | 2023 |
| • Computer Science and Engineering Fellowship (merit-based doctoral fellowship), UC San Diego | 2023 |

TEACHING & SERVICE

- Reviewer: CVPR 2025, NeurIPS 2025, CVPR 2026, NeurIPS 2026, ICLR 2027.

Head TA, Machine Learning, Sharif University of Technology (SUT)

Feb. 2021 – Aug. 2021